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The Domestic Chicken (*Gallus gallus*) in Sub-Saharan Africa: A Background to its Introduction and its Osteological Differentiation from Indigenous Fowls (Numidinae and *Francolinus* sp.)*

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The co-occurrence of domestic guineafowl (*Numida meleagris*) and chicken (*Gallus gallus*) in prehistoric sites both in Africa and Europe has necessitated the construction of an osteological key to aid in their differentiation. This study deals particularly with the osteomorphological and to a lesser extent osteometric separation of fowls in a West African context (including larger *Francolinus* sp., *Numida, Guttera* and *Agelastes* sp. and *Gallus gallus*). This article will, however, be of interest to other Africanist and European researchers as well. The spread of the African guineafowl into Europe during the 5th century BC or earlier and Asiatic chickens into West Africa before 850 AD are important indicators of cultural contact which have gone relatively unnoticed. A summary of archaeological data available concerning the chicken in Africa is provided to give a background to the problem.

Keywords: SUB-SAHARAN AFRICA, DOMESTIC CHICKEN, GUINEAFOWL, FAUNAL REMAINS, OSTEOMORPHOLOGY, JENNE-JENO.

Introduction

The origin and spread of the domesticated chicken (*Gallus gallus*) from its southeast Asian homeland throughout Europe has been well documented by Barbara West & Ben-Xiong Zhou (1988) in their article "Did Chickens Go North? New Evidence for Domestication." This article did not, however, deal with the dispersion of the chicken into Africa. Nor has any publication, of which the author is aware, dealt with the diffusion of the domestic helmeted guineafowl or pintade (*Numida meleagris*) from Africa into Europe—a process which began at least as early as the fifth century BC (Mongin & Plouzeau, 1984). Thus, many co-occurrences of guineafowl (*Numida meleagris*) and chicken (*Gallus gallus*) in

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archaeological sites both in Europe and Africa, are likely to have long gone effectively unobserved in the archaeozoological record. While the general osteological similarity of these two species has no doubt contributed to the confusion, the gap in knowledge is more attributable to a lack of informed research and readily available criteria for the separation of these taxa. Beyond their importance as markers of trade or cultural contact, the positive identification of *Gallus gallus* and *Numida meleagris* in archaeological contexts will allow for a better understanding of the development and exploitation of these two species over space and time.

The author first became involved in the problem of these birds' differentiation while analysing avian remains from the West African Iron Age site of Jenne-jeno (Inland Niger Delta, Mali, Figure 1) (MacDonald, in press; cf. McIntosh & McIntosh, 1980). The presence of anomalous, *Gallus*-sized spurred tarsometatarsi in that assemblage (Figure 2) made it evident that many elements previously assigned to Numidinae might indeed be *Gallus* (there being no spurred birds of an equivalent size present indigenously in West Africa, only smaller birds such as francolins and slightly spurred, deep-forest dwelling *Agelastes* sp.).

Subsequent investigation proved the presence of *Gallus gallus* throughout the latter two phases of the Jenne-jeno assemblage (Phase III *c.* 450–850 AD and Phase IV *c.* 850–1400 AD) (cf. MacDonald, in press). Very recently, the author has also identified *Gallus* remains from early contexts at other Malian Iron Age sites excavated and surveyed in 1990 by Téréba Togola, Helen Haskell and himself. Numerous bones from the early and middle Phase IV layers of the occupational mound of Akumbu A (*c.* 1000–1280 AD and *c.* 1280–1400 AD) have been assigned positively to *Gallus*. Additionally at Toladié, located like Akumbu in the dead delta of the Niger River (aka the Mema region, Figure 1), *Gallus* remains have been identified by the author eroding out of gullies associated with Phase III and IV ceramics. In some cases these remains were so diminutive in size (comparable to modern bantam breeds or wild junglefowl) that their morphological differentiation from smaller indigenous West African fowls—such as larger francolin species, became necessary (see below).

Archaeological Background

The presence of chicken at Jenne-jeno before 850 AD is worthy of further comment in three respects: its early date, its small size and the lack of hypotheses presently formulated to deal with these issues. Sadly, questions regarding the introduction of *Gallus gallus* into Africa and its subsequent dispersion have been neglected in Africa's archaeological studies.

The earliest osteological evidence in proximity to Africa is from the site of Sweyhat in northern Syria dating to 2400 BC (Buitenhuis, 1983). The earliest known skeletal remains from Africa itself are from Egypt and date to the eighteenth dynasty (*c.* 1567–1320 BC) (Darby *et al.*, 1977). The earliest artistic evidence is that of a painted limestone fragment depicting a "red junglefowl" recovered during excavations in the Valley of the Kings in an "undisturbed stratum between the tomb of Ramses IX and the Eighteenth Dynasty tomb-chamber wherein the cache of Ikhnaton (Amenophis IV) was made . . ." (*c.* 1425–1123 BC) (Carter, 1923: 1). It is interesting to note that this find was misrepresented by Zeuner (1963: 445) as being a potsherd from the tomb of Tutankhamun from *c.* 1350 BC. The earliest literary evidence of the chicken in Africa is a record in the Annals of King Thutmose III (Eighteenth dynasty, *c.* 1504–1450 BC) of four gift fowls from Babylonia (Darby *et al.*, 1977; Carter, 1923). It must be asserted, however, that the chicken did not become a common bird in the Egyptian subsistence regime until almost a thousand years after these incidents (von den Driesch, pers. comm.; Darby *et al.*, 1977).

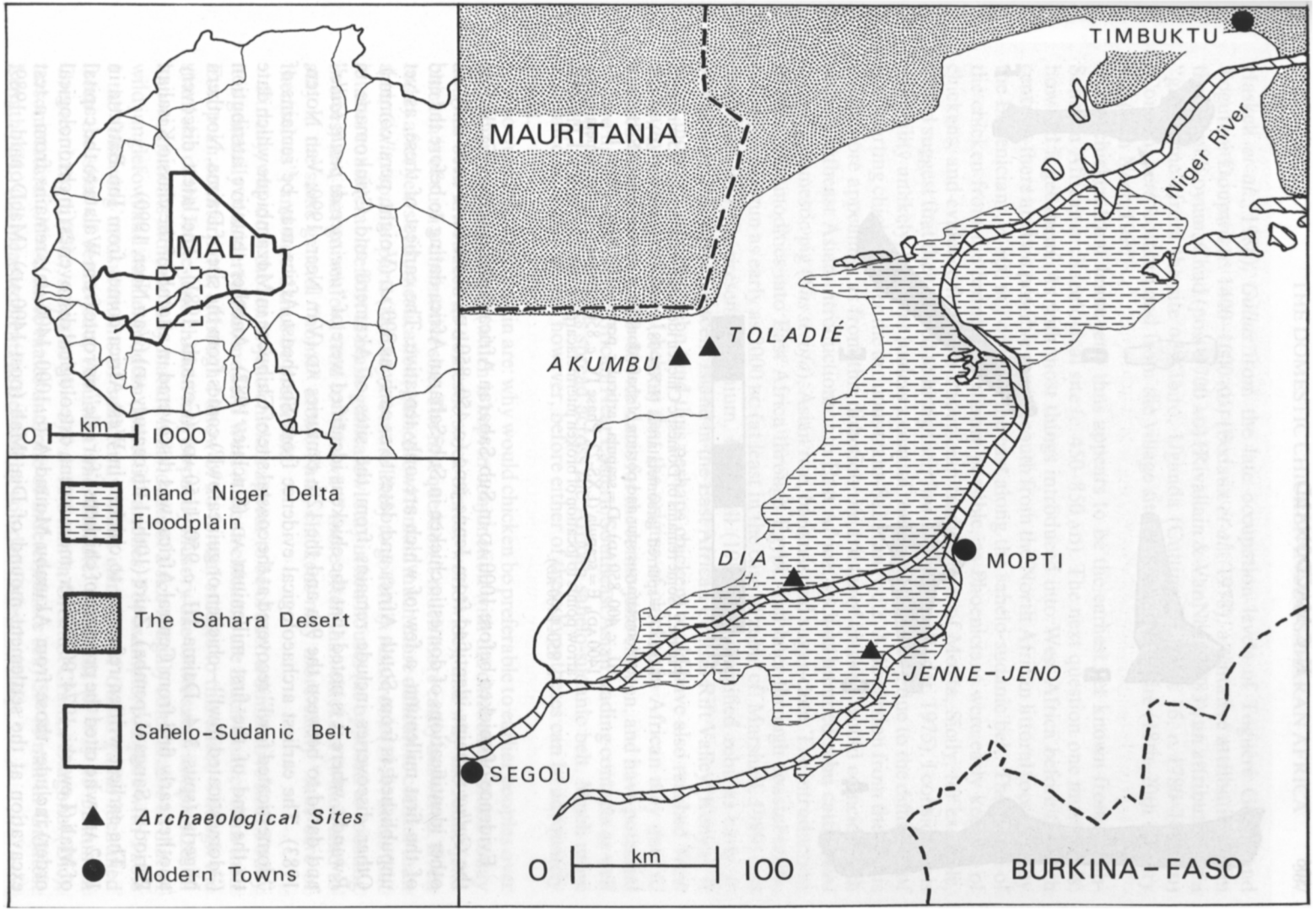


Figure 1. Archaeological sites in the Inland Niger Delta and Mema regions of Mali, West Africa where remains of Domestic Chicken (*Gallus gallus*) have been identified by the author (illustration by K. C. MacDonald and G. N. Bailey).

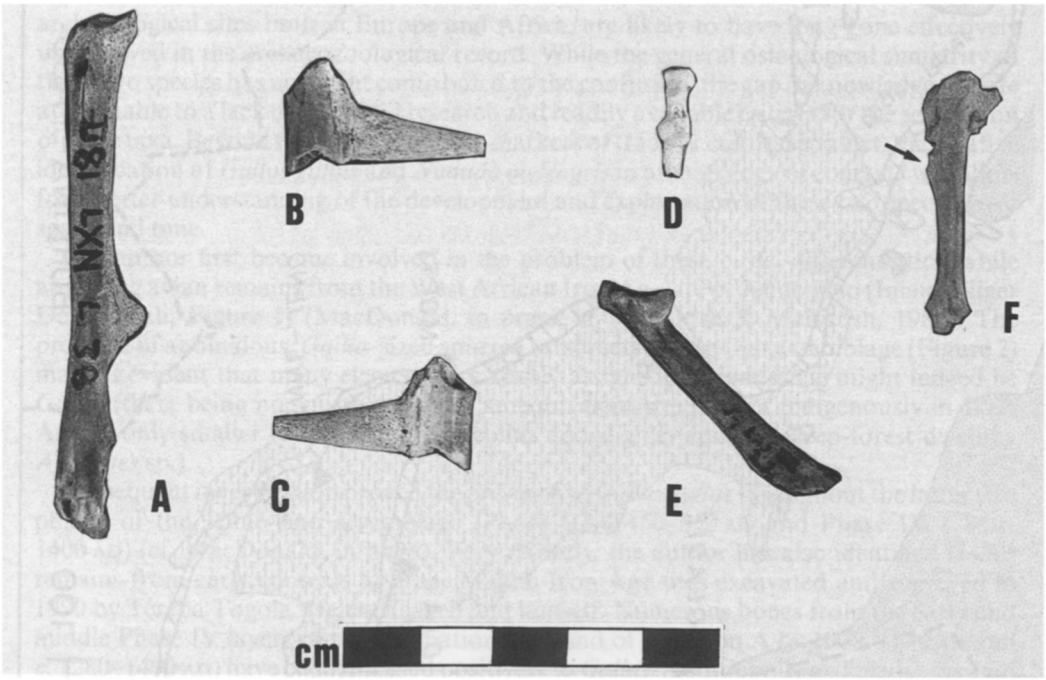


Figure 2. Some remains of Domestic Chicken identified from the site of Jenne-jeno, Mali. A = spurred tarsometatarsus (LXN-38, Phase III, *c.* 400–850 AD), B–C = tarsometatarsus spur and portion of shaft, dorsal and planar view (LXS-71, Phase III, *c.* 400–850 AD), D = scapula, articular portion (LXN-18, Phase IV, *c.* 850–1400 AD), E = scapula (LXS-22, Phase IV, *c.* 850–1400 AD), F = carpometacarpus, arrow points to location of broken intermetacarpal tuberosity (Mnd 2–1, Phase IV, *c.* 850–1400 AD).

Evidence of chicken before 1000 AD in Sub-Saharan Africa remains scarce. Excluding the *Gallus* remains identified from Jenne-jeno (*c.* 450–850 AD), there have been only six other identifications of domestic chicken in Sub-Saharan Africa dating to before the end of the first millenium, a few of which are only tentative. The earliest of these, as yet unpublished, is from South Africa and dates to as early as 800 AD (Voight, pers. comm.). Other discoveries include remains from the sites of Akameru and Cyinkomane in Rwanda, where it is noted that the chickens identified were of "*une race de petite taille*" and dated to between the 9th and the 12th centuries AD (Van Neer, 1990; Van Noten, 1983). The earliest archaeological evidence from Southeast Africa may be remains of "domesticated fowl" recovered at the coastal site of Chibuene in Mozambique which date to the end of the first millenium AD (Sinclair, 1982). Another tentative attribution ("domesticated fowl"—chicken or guineafowl?) comes from the site of Daima, Northern Nigeria (spits 3–4, Daima III, *c.* 950–1150 AD) (Connah, 1981). The latest discovery another early find from Central Africa, was discovered in a tomb of the classic Kisalian Period at Sanga (Upemba), Zaire (10th–13th century AD) (Van Neer, 1990).

The earliest written reference to chicken in West Africa comes from Ibn Battutta in 1352 AD, who cited the presence of chickens "for sale" en route from Walata to the capital of Mali (Lewicki, 1974: 90). Other, more recent, osteological discoveries (in chronological order) include: those from Akumbu Mound A (*c.* 1000–1400 AD), remains from a test excavation at the settlement mound of Dia, Mali (post-1400 AD) (MacDonald, 1989;

Haskell *et al.*, 1988), *Gallus* from the late occupation levels of Toguéré Gailia and Toguréré Doupwil (*c.* 1400–1600 AD) (Bedaux *et al.*, 1978), a tentative attribution from the site of Koyum, Chad (post-1700 AD) (Rivallain & VanNeer, 1983), an attribution of a "galliforme" from the site of Kibiro, Uganda (Cutting III, Unit 16, *c.* 1780–1880 AD) (Connah, pers. comm.) and from the village site of Kiantapo, Zaire (18th–20th century AD) (Van Neer, 1990).

The chicken from Jenne-jeno thus appears to be the earliest yet known from a Sub-Saharan African archaeological site (*c.* 450–850 AD). The next question one must ask is: how did it get there? As with most things introduced into West Africa before the 16th century, there are two possible routes: south from the North African littoral (occupied by the Phoenicians, *c.* 800 BC) or from the east along the sahelo-sudanic belt. The origin of the chicken from the former is certainly possible, as Phoenicians were early keepers of chickens, and evidence from the Phoenician trading port of Moyta, Sicily, for example, would suggest that they tended to bring their fowl with them (Ryder, 1975). I consider this possibility unlikely, however (unless a coastal route was followed), due to the difficulty of transporting chickens over the arduous trans-Saharan route. A dispersion from the east is much more appealing, as from this direction the arrival of chicken would coincide with other Southeast Asian introductions into East Africa, such as humped zebu cattle (*Bos indicus*), domestic pig (*Sus scrofa*), Asian rice, bananas and sugar-cane. The introduction of these commodities into East Africa through Egypt, and later through coastal trade, could have begun as early as 2000 BC (at least in the case of zebu, cf. Marshall, 1989) and continued into the present millenium. Marshall (1989) has identified zebu as early as 500 BC from the site of Crescent Island in the East African Central Rift Valley (Kenya). It is not inconceivable that zebu (and thus perhaps chicken) could have also reached West Africa by that date. Thus, I believe that the origins of chicken in African may date to appreciably before the *c.* 500–800 AD dates for the Jenne-jeno chicken, and have potential to shed light on early East African/Near Eastern/Southeast Asian trading contacts as well as early relations of peoples spread across the African sahelo-sudanic belt. Much more excavated data are needed, however, before either of these hypotheses can be adequately tested.

Two questions that remain are: why would chicken be preferable to earlier peoples over the indigenous guineafowl (which is kept in conjunction with the chicken in Africa today and has been introduced as domestic stock almost worldwide), and why the Jenne-jeno chicken is of such a diminutive size. The first question is much more easily answered. Even today after years of domestication, guineafowl remain much more difficult to control than their *Gallus* counterparts (i.e. they are more likely to go feral). Ashbrook (1951) notes that it takes considerable effort to make guineafowl stay in hen houses (as opposed to roosting in trees) and that guineafowl tend to "fly away" more often than chickens (which tend to stay in the same general "home" area). Thus, chickens may more easily "take care of themselves", and would thus be preferable.

Finally, why does the Jenne-jeno chicken appear to be of such a diminutive size? Today it is generally considered that modern West African chickens are much smaller than their European counterparts, although I know of no formal studies to this effect. Though the small size of the Jenne-jeno chicken may be partly illusory due to the small sample size, all known elements indicate fully mature individuals of a size roughly comparable to that of wild junglefowl and smaller modern domestic varieties. For example, compare with Table 1 the following measurements of sufficiently intact elements of *Gallus gallus* identified from the sites of Jenne-jeno, Akumbu and Toladié: coracoids ($N=3$) 45–52 mm in Greatest Length (GL), scapulae ($N=5$) 9·6–10·6 mm in Greatest Cranial Diagonal (Dic), tarsometatarsi ($N=4$: three males and one of indeterminate sex) 59–70 mm in Greatest Length (GL) (all measurements as defined by von den Driesch, 1976).

Other skeletal elements of *Gallus* recovered from these Malian Iron Age sites show a similarly diminutive trend. To this end, it is interesting to note that domestic *Gallus* remains from Halstatt/La Tene contexts in Iron Age Europe are also very small compared with modern breeds (Thesing, 1977, Table 1). This trend continued in Europe until early Roman times when the size of European chickens both within and outside the Roman Empire increased to approximately modern levels. The reasons for this sudden increase, and for a subsequent brief dip in the size of European chickens after the fall of the Roman Empire, remain relatively unexplored (Thesing, 1977; von den Dreisch, pers. comm.). The author has suggested that at least in Africa, diminutive size is probably linked with chickens being left to forage for their own food by their keepers—a practice which continues in West Africa to this day (MacDonald, in press). This "feral" activity, coupled with environmental factors and possible encouragement on the part of breeders of sturdier birds, would have kept the African chicken close to the size of early domesticates or wild junglefowl. At thus time, however, we have insufficient identified *Gallus* remains over space and time in Africa to investigate these issues.

Osteological Differentiation

To further this study on the origins and development of the chicken in Africa, the following criteria have been established for the differentiation of *Gallus* from potentially confusing African fowls: the guineafowl (Numidinae: wild savannah and forest genera, and domestic varieties) and larger francolin species (i.e. *bicalcaratus, icterorhynchus, clappertoni, capensis, jacksoni, castaneicollis, erkelii, swainsonii, rufopictus* and *leucoscepis*). It should be noted that the osteomorphological nomenclature used is that of Baumel (1979), or, where appropriate, Gilbert (1985).

The genus *Francolinus* as a member of Gallinae shares in common many morphological characteristics with *Gallus*, although surprisingly in some respects (the scapula and the radius) it conforms osteomorphologically with Numidinae. While a few elements of larger *Francolinus* sp. may be positively separated on the basis of morphology from *Gallus* (the coracoid, the scapula, the pelvis, the radius and the tarsometatarsus), most are undifferentiable in cases of extralimitally small domestic chickens. Such cases are rare, however, and *Gallus* may be usually satisfactorily differentiated from *Francolinus* on the basis of metrics (Table 1) and/or associated diagnostic elements.

Numidinae is a sub-family composed of four genera and six species: *Agelastes* (two species), *Guttera* (two species), *Acryllium* (one species), and *Numida* (one species). The most important of these taxa, *Numida meleagris* (the wild or domestic helmeted guineafowl), today occurs almost as globally as the chicken itself. Wild varieties continue to exist throughout Sub-Saharan Africa. The remaining two genera (*Agelastes* and *Guttera*—the Plumed and Crested, and the Black and White-breasted Guineafowls, respectively) are deep-forest dwelling varieties today distributed patchily in West, Central and south-eastern Africa. Osteologically (in all cases but the cranium and the tarsometatarsus), the positive differentiation of *Numida, Agelastes* and *Guttera* is not possible. This need not concern us, however, as our predominant concern is to separate Numidinae and larger *Francolinus* sp. from *Gallus gallus*.

Of the genera mentioned above, only the genus *Acryllium* is not treated here as it occurs only in the horn of East Africa—outside of the original study range of West Africa. A cursory study of remains available at the Field Museum of Natural History and the U.S. National Museum of Natural History demonstrates that *Acryllium* and *Agelastes* and *Guttera* conform in most ways with *Numida* osteologically. Obvious exceptions include: the mature tarsometatarsus of *Acryllium*, which has a number of shallow, broad spurs giving it a roughly "serrated" appearance, the cranium of *Acryllium* which does not possess a bony casque as *Numida*, and the pelvis of *Acryllium* in which the process

Table 1. Osteological size ranges in comparative specimens studied and from published work

	Numida meleagris	*Gallus gallus*	*Gallus gallus*	*Gallus gallus*	*Francolinus bicalcaratus*
Remarks	Wild and domestic	Modern domestic	Hallstatt (Thesing, 1977)	Junglefowl	—
Number of specimens	8 (4 male, 4 female)	10 (4 male, 6 female)	Variable: 9–22 (indeterminate)	4 (2 male, 2 female)	3 (2 male, 1 female)
Cranium-GB	27·5–31·6	23·2–34·7	na	24·8–25·8	23·5–27·3
Mandibular-LaF	41·7–50·2	34·3–64·8	na	38·4–46·6	38·2–44·9
Sternum-LC	89·1–105·6	62·0–104·2	na	68·9–89·3	58·1–73·5
Sternum-SBF	20·9–25·8	20·5–41·0	na	22·2–26·9	18·0–20·8
Coracoid-GL	51·8–57·9	40·1–72·0	41·0–63·7	42·8–50·4	40·3–45·5
Scapula-Dic	12·7–15·2	9·3–15·5	na	10·0–11·3	8·8–11·1
Humerus-GL	72·2–83·4	48·2–91·2	59·0–70·0	57·5–69·7	53·6–59·9
Humerus-BP	19·7–22·9	13·7–24·6	16·0–20·0	14·5–18·0	13·7–16·0
Humerus-BD	16·1–17·9	11·0–19·2	13·0–15·0	12·1–14·5	10·6–11·7
Radius-BD	6·8–8·1	5·4–9·7	na	5·1–6·7	5·0–5·3
Ulna-Dip	13·2–15·0	9·8–16·5	na	10·1–11·9	8·3–10·8
Carpometacarpus-GL	40·0–44·0	23·9–48·8	na	32·3–37·1	27·8–30·8
Carpometacarpus-BP	11·6–13·1	8·9–13·9	na	9·9–11·2	8·0–8·8
Phalanx 1-GL	14·1–16·0	10·6–16·5	na	11·5–13·1	10·2–11·8
Pelvis-LV	74·5–85·4	47·9–83·4	na	58·1–76·0	50·0–57·0
Pelvis-DiA	6·8–8·8	5·5–11·9	na	5·6–6·5	5·2–5·8
Femur-GL	80·6–83·4	48·7–114·6	63·8–79·6	62·6–75·8	60·0–72·5
Femur-BP	12·7–19·8	12·6–22·3	na	12·5–15·0	11·9–14·6
Femur-BD	10·7–17·2	11·1–21·2	12·0–15·8	11·1–13·8	9·8–11·8
Tibiotarsus-GL	112·8–122·0	65·6–156·4	84·8–112·7	89·3–114·0	78·4–91·6
Tibiotarsus-Dip	20·5–23·3	13·8–26·0	na	15·3–18·3	11·0–14·0
Tibiotarsus-BD	10·9–12·9	8·9–14·8	8·3–14·5	8·7–10·6	7·9–10·3
Tarsometatarsus-GL	70·6–81·3	40·8–114·4	61·0–75·5	59·0–73·3	51·6–62·6
Tarsometatarsus-BP	12·4–14·3	10·5–17·7	12·3–13·3	9·8–12·0	9·5–10·4
Tarsometatarsus-BD	12·4–14·9	9·3–19·3	12·8–13·2	10·6–11·9	8·2–10·2

All measurements are in mm and follow the von den Dreisch system (1976).

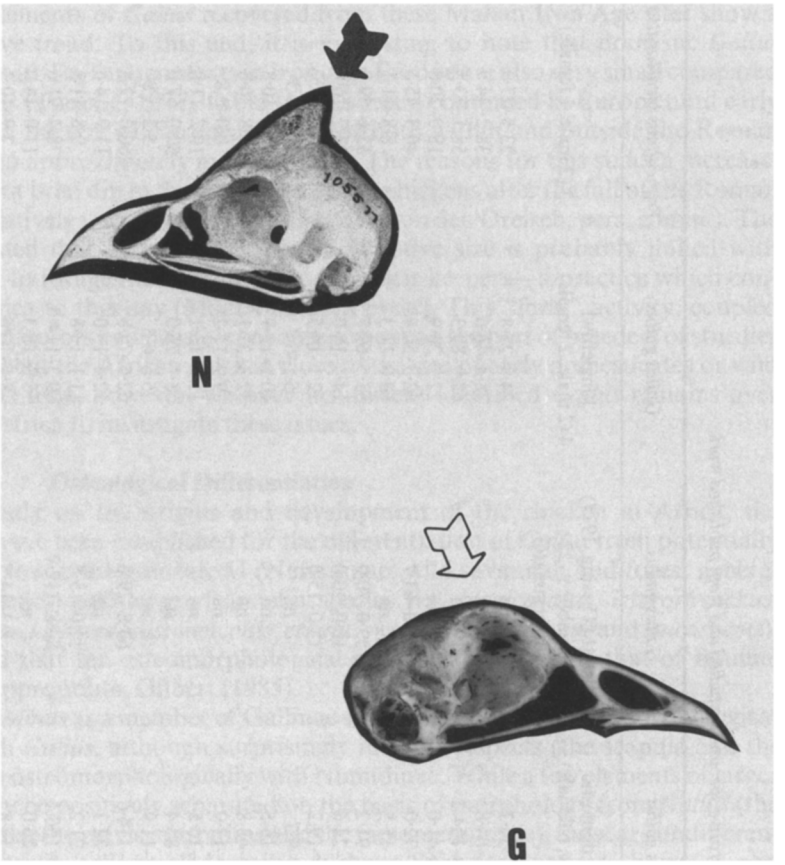


Figure 3. Differentiation of the Cranium, N = *Numida*, G = *Gallus*.

terminalis ilii is even more elongated than that of *Gallus*. Until a formal study can be performed on this species, however, researchers working in regions where *Acryllium* occurs (or may have occurred) should consult reference specimens.

The important diagnostic characteristics of each element are as follows:

(i) The cranium (Figure 3)

The cranium of *Numida meleagris* is uniquely distinguished by a bony casque or "helmet" on the dorsal surface of its skull. While the casque is more pronounced on mature male individuals, it is discernible on females and immature individuals. No other members of the sub-families Numidinae or Gallinae possess such a characteristic.

(ii) The scapula (Figure 4)

The cranial extremity of the scapula can be used to differentiate positively between *Gallus, Francolinus* and Numidinae. In Numidinae and *Francolinus* the acromion process is strongly mesially projected. In *Gallus* the acromion process is compact and does not project. Scapulae of *Francolinus* and *Numida* may be differentiated metrically by the Greatest Cranial Diagonal (Dic) measurement (Table 1).

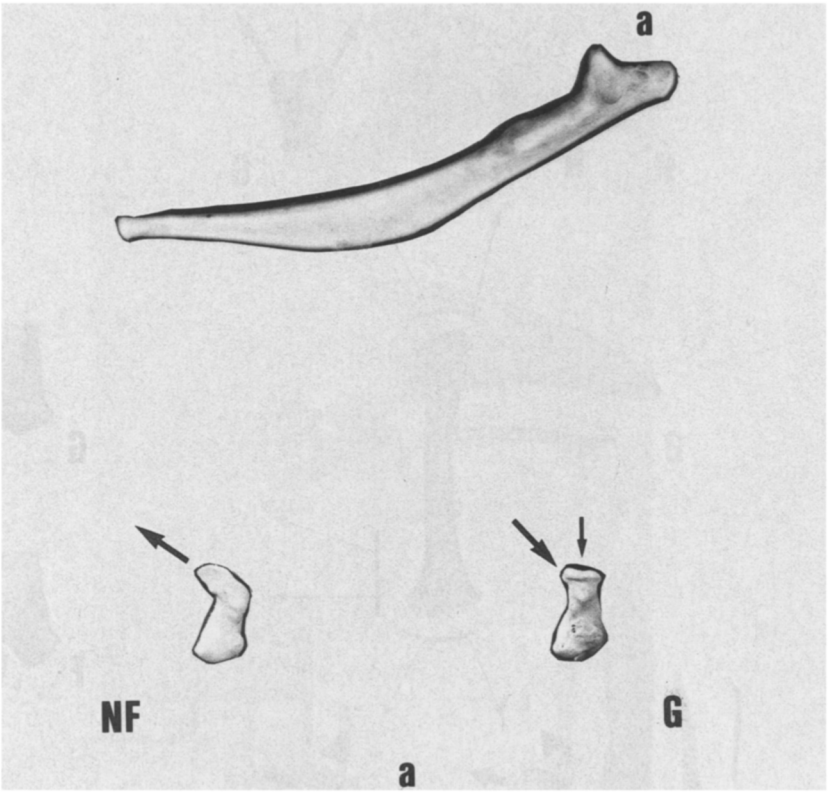


Figure 4. Differentiation of the Scapula, NF = Numidinae/*Francolinus*, G = *Gallus*.

(iii) The coracoid (Figure 5)

The coracoid may be used to differentiate between Numidinae and Gallinae at two points, and between *Gallus* and *Francolinus* at one point.

On the proximal surface of the coracoid, the facies articularis clavicularis are more widely separated in Gallinae than in Numidinae. Typically, there is a slight sulcus present between the facies at their mesial extremity in Gallinae. Note that there is a degree of gradation in this trait, making it unreliable in marginal cases.

On the distal end of the coracoid, the facies articularis sternalis is constricted superio-inferiorly and extended mesio-laterally in Gallinae. In Numidinae this facies is expanded superio-inferiorly (especially at its mesial extremity) and is compact mesio-laterally. *Gallus* may be positively differentiated from *Francolinus* by the marked concavity of the facies articularis sternalis of *Francolinus*. That of *Gallus* is only midly concave.

(iv) The sternum (Figure 6)

The sternum, if recovered at least partially intact, is very diagnostic between Gallinae and Numidinae.

First, the cranio-lateral process is long with a predominantly cranial projection in Gallinae, while short and blunt with a weakly lateral projection in Numidinae.

Second, the fenestra lateralis of Gallinae extends cranially to a point just caudal of the facies articularis costalis. In Numidinae this fenestra ends more caudally, approximately at the mid-point of the carina sterni.

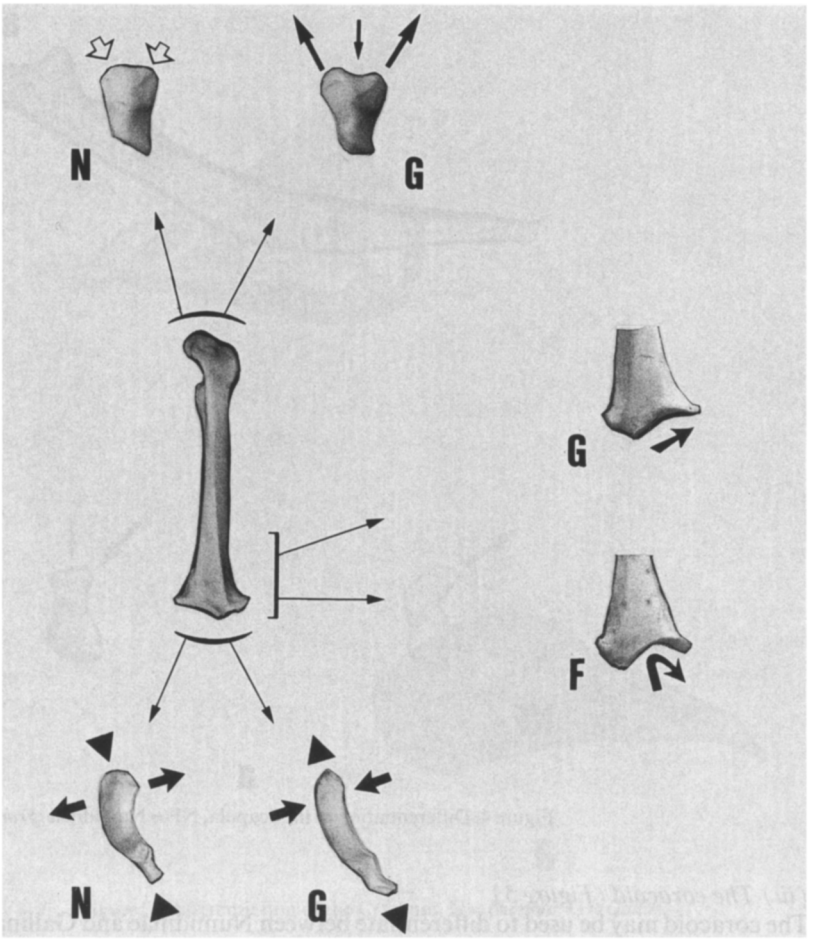


Figure 5. Differentiation of the Coracoid, N = Numidinae, G = *Gallus*, F = *Francolinus*.

(v) The pelvis (Figures 7 and 8)

As with the sternum, the pelvis when recovered semi-intact is very diagnostic. In fact, even more so, as it may be used to differentiate *Francolinus* and *Gallus*.

The ala postacetabularis ilii or postacetabular wing, is extended caudally in *Gallus* far beyond the synsacrum with an elongated and strong process terminalis ilii. In Numidinae the ala postacetabularis ilii is truncated just caudally of the last synsacral vertebrae and the process terminalis ilii is relatively weak. In *Francolinus* the ala postacetabularis ilii mimics *Gallus* in its projection, but is more triangular in shape than that of *Gallus*, which is rectangular.

The process terminalis ischii of *Francolinus* are caudally projected and extend caudally far beyond the process terminalis ilii. In comparison those of *Gallus* and Numidinae are not projected and end at a point only just caudal of the process terminalis ilii.

(vi) The radius (Figure 9)

The radius may be used to separate *Gallus, Francolinus*, and Numidinae. The Styloid Process on the distal end of the radius is usually more laterally projected and has a more

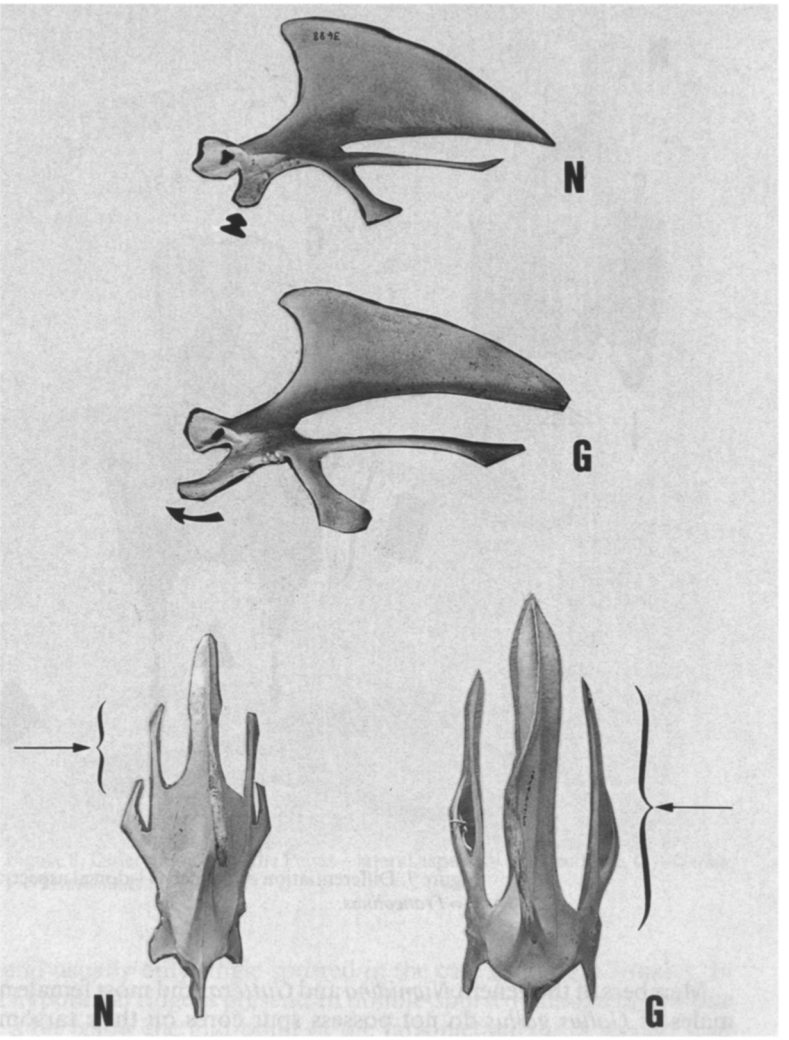


Figure 6. Differentiation of the sternum, N = Numidinae, G = Gallinae.

clearly defined apex in Numidinae and *Francolinus* than in *Gallus* (see also radius Distal Breadth BD] measurements for all taxa in Table 1 to separate *Francolinus* from Numidinae).

(vii) The carpometacarpus (Figure 10)
The carpometacarpus may be readily used to separate Gallinae and Numidinae. Gallinae possess a highly developed intermetacarpal tuberosity. Numidinae do not possess an intermetacarpal tuberosity.

(viii) The tarsometatarsus (Figure 11)
Tarsometatarsi can be utilized to differentiate most individuals of the taxa concerned here. While the morphologies of the articular surface of the tarsometatarsus in Gallinae and Numidinae are practically indistinguishable, their spurs are highly distinctive.

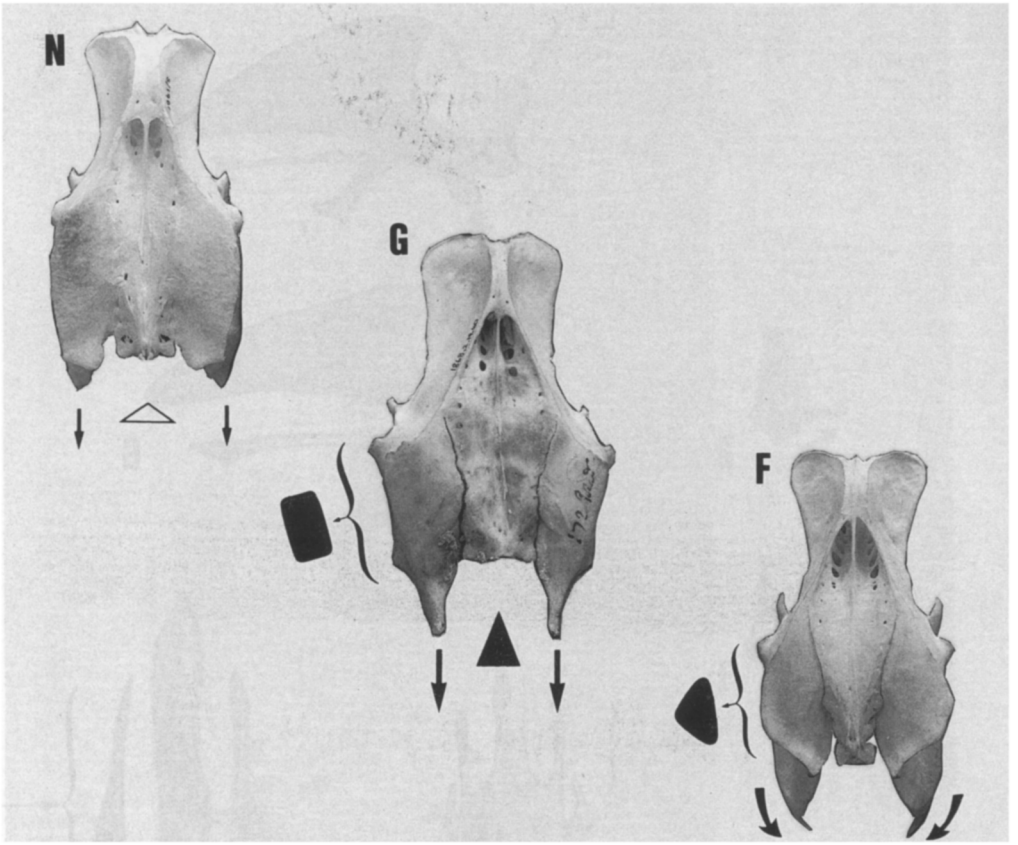


Figure 7. Differentiation of the Pelvis—dorsal aspect, N = Numidinae, G = *Gallus*, F = *Francolinus*.

Members of the genera *Numidina* and *Guttera*, and most female members and immature males of *Gallus gallus* do not possess spur cores on their tarsometatarsi. Mature male members and some mature female members of *Gallus gallus*, and all mature members of the genus *Agelastes* and the genus *Francolinus* have spurred tarsometatarsi. Spurred tarsometatarsi of *Gallus gallus* may be differentiated from those of these other fowls by two primary traits: the size of the "spur-core" itself and the presence or absence of a pronounced and extended hypotarsal ridge with or without secondary spur.

Wild and domesticated *Gallus gallus* spur-cores, depending upon breed and maturity, may extend, in the author's experience, up to *c.* 40 mm from articulation with the shaft to the core's apex. Those of *Francolinus* are rarely more than 10 mm in length, and no more than 19 mm in extreme cases. While there are few osteological comparative specimens of *Agelastes* in the world's museums, measurements of the spurs of captured specimens (usually overestimates due to their inclusion of the spur-core's keratinous covering) range from 5–8·5 mm (cf. Urban *et al.*, 1986).

Gallus gallus is the only spurred taxon under consideration lacking an extended hypotarsal ridge. This ridge is pronounced in both the genus *Agelastes* and the genus *Francolinus* and extends inferiorly to the spur-core in *Agelastes* and past the basal spur-core in *Francolinus*. The tarsometatarsi of *Francolinus* are usually double-spurred in the

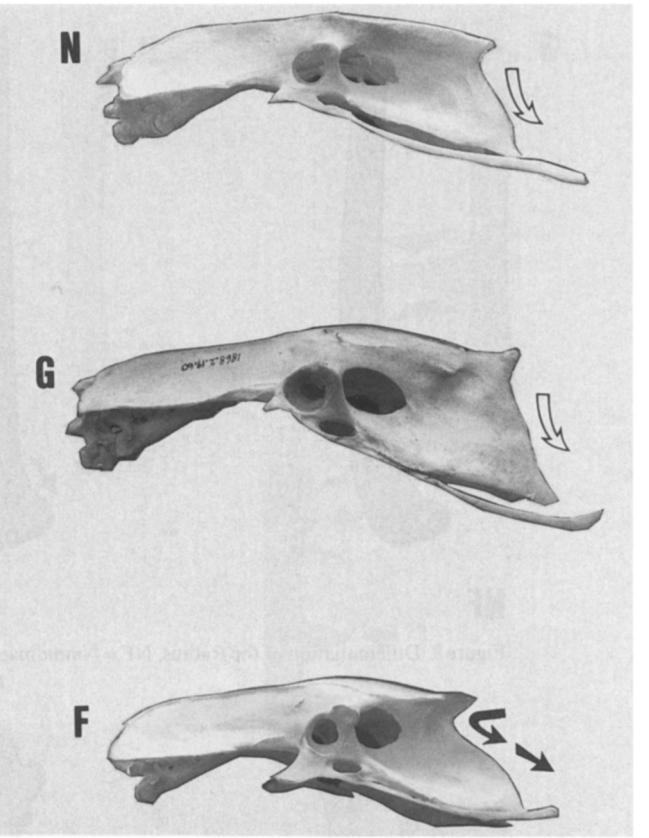


Figure 8. Differentiation of the Pelvis—lateral aspect, N = Numidinae, G = *Gallus*, F = *Francolinus*.

case of mature males, and usually only single-spurred in the case of mature females. In both cases an extended hypotarsal ridge is present. In double-spurred tarsometatarsi one spur occurs as normal a bit below the mid-point of the tarsometatarsus, a second spur (usually smaller than the first) occurs on the hypotarsal ridge near the proximal end of the bone.

It should be remembered that except where noted the genera in question conform with the general morphological characteristics of their particular sub-families—Numidinae and Gallinae. Additionally, I would like to stress that while I have found numerous "secondary morphological traits", on elements mentioned and not mentioned above, I have represented here only those non-metric traits which I feel are reliable enough in themselves to make a definite taxonomic assignment.

Conclusions

It is known that the chicken first arrived in Sub-Saharan Africa sometime before 850 AD and that guineafowl were first imported into Europe during the 5th century BC (Mongin & Plouzeau, 1984). It is hoped that the criteria above, when commonly utilized in avifaunal analysis, will bring about a greater knowledge of the dispersion of these two birds into Africa and Europe—with all of their subsequent implications for trade and cultural

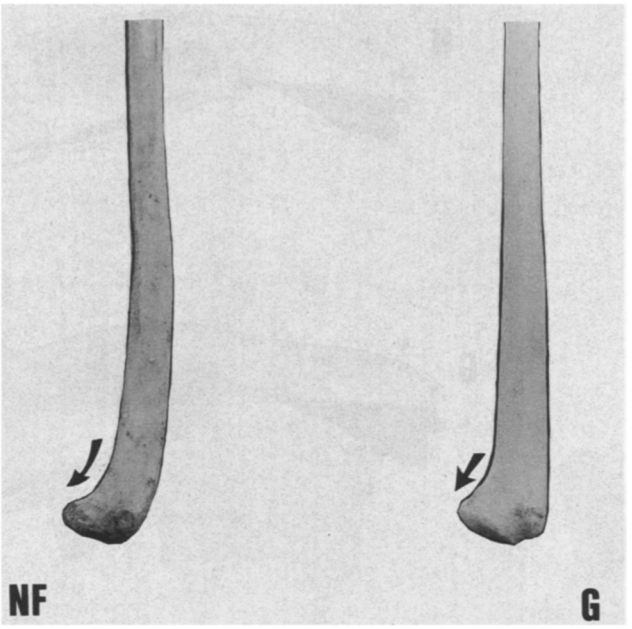


Figure 9. Differentiation of the Radius, NF = Numidinae/*Francolinus*, G = *Gallus*.

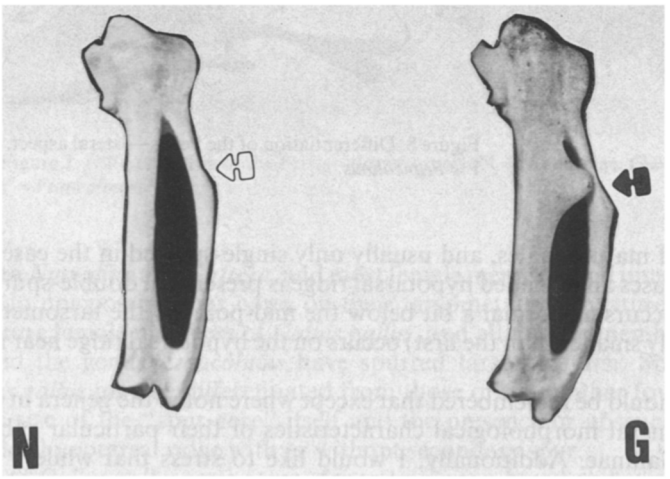


Figure 10. Differentiation of the carpometacarpus, N = Numidinae, G = Gallinae.

contact. Perhaps more importantly from an archaeozoological standpoint, however, an increased pool of osteometric data may shed light on the development of domesticated fowl in Sub-Saharan Africa.

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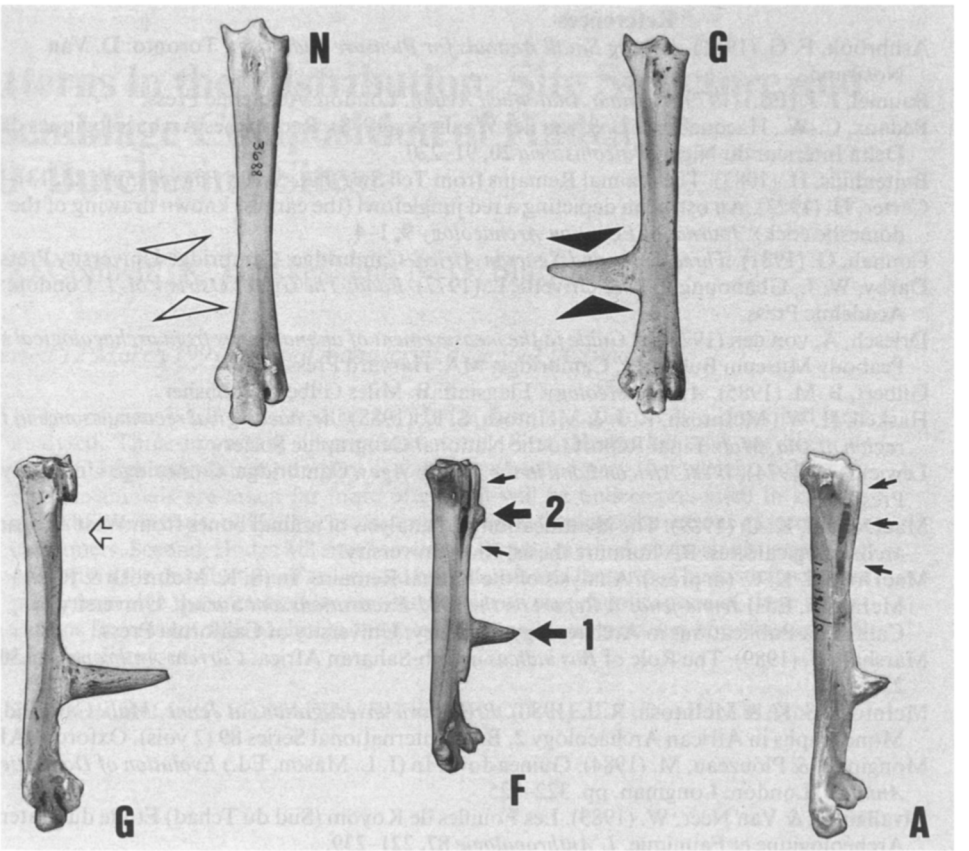


Figure 11. Differentiation of the tarsometatarsus, N = *Numida* and *Guttera*, G = *Gallus*, F = *Francolinus*, A = *Agelastes*.

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